

There are three lessons for an asset owner:

1. Many assets can lose money in the short run.

This is the standard channel of how we should evaluate risk.

2. What appears safe in the short run can, in fact, be quite risky over the long run.

The market cares about long-run risk, and small adjustments made today can result in large effects in twenty to thirty years time. In the context of factor model theory, assets with large exposures to long-run risk (high long-run risk betas) need to have high returns. It turns out that assets with high long-run risk betas include value stocks and other popular investment strategies such as currency carry trades.¹¹ How much long-run risk can you tolerate as an investor?

3. Volatility as a risk factor matters.

In Bansal–Yaron, it is consumption volatility, but in a more general setting, it is macro volatility. Volatility risk is different from consumption (macro) risk. It carries its own risk premium. We discussed volatility as a risk factor in chapter 7.

3.4. HETEROGENEOUS INVESTORS

One criticism of Mehra–Prescott involved the assumption of the representative agent. The world has *heterogeneous* investors, not a single representative agent.¹² Perhaps the representative agent isn't really representative. This was first shown by Jerison (1984), in a paper that never got published but was expanded and extended by Kirman (1992). Jerison and Kirman show that in some circumstances the preferences of the representative agent are not the (weighted) average of preferences of individual agents in the economy. To be blunt, if all agents in the economy prefer bananas over apples, it is possible to construct a representative agent who prefers apples over bananas. Furthermore, how that representative agent feels about apples versus bananas in different states of the world leads to the same prices as the economy where individual agents optimize and prefer bananas over apples. Thus, rejecting a representative agent model with implausibly high risk aversion may not say anything about how a true heterogeneous agent economy works.

The profession remains wedded to representative agent models, though. Some of the reasons are historical and due to tractability; representative agent models are much easier to solve than heterogeneous agent models. But one good reason is that, in a world full of heterogeneous agents, there is still economic meaning for a representative agent even though he may not be the “average” agent, where the average is a simple average or even an average taken over wealth. In fact, some

¹¹ See Bansal, Dittmar, and Lundblad (2005) and Bansal and Shaliastovich (2010), respectively.

¹² Heterogeneous agent models were first examined by Bewley (1977).